

Question description

A beautiful *code* of a tree of n nodes is a sequence of $n-2$ integers that uniquely specifies the structure of the tree.

The code is constructed as follows: As long as there are at least three nodes left, find a leaf with the smallest label, add the label of its only neighbour to the code, and remove the leaf from the tree.

Given a beautiful code of a tree, your task is to construct the original tree.

Constraints

- $3 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Input

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

The second line contains $n-2$ integers: the beautiful code.

Output

Print $n-1$ lines describing the edges of the tree. Each line has to contain two integers a and b : there is an edge between nodes a and b . You can print the edges in any order.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5
2 2 4

EXPECTED OUTPUT

1 2
3 2
2 4
4 5

Test Case 2

INPUT (STDIN)

5
3 1 2

EXPECTED OUTPUT

4 3
3 1
1 2
2 5

▼ Mandatory Test Cases

Test Case 1

KEYWORD

q.pop();

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

300

Test Case 3

NLOC

34